Submission Highlights: Master Simulation Suite

Target Journal: Fusion Engineering and Design (Elsevier) | Author: Jacob James Rasmussen

- **Fast-neutron cascades contract tungsten CPS pore radius from 25.0 to 15.2 um.**
- **Carman-Kozeny permeability exhibits a 70% non-linear decay under 20 dpa flux.**
- **Critical dryout heat flux drops to 22.5 MW/m^2 at 9.8 dpa (SF = 1.50 limit).**
- **Complete thermal dryout parity with baseline heat flux occurs at 13.0 dpa.**
- **Spacetime Algebra Cl 1,3 rotors preserve real energy positivity T00 >= 0.**
- **Turn-to-turn contact resistivity of 15.0 uOhm.cm2 shunts radial quench currents.**
- **Passive current decay bounds peak quench temperatures at T_max <= 200 K.**
- **3D anisotropic Tsai-Wu yield metric f_TW = 0.244 <= 1.0 under disruption loads.**
- **Rotating n = 1 halo currents (1.8 MA, 120 Hz) produce peak DAF = 1.85.**
- **Inconel 718 keyway interlocks achieve Strength Reserve Ratio SRR = 1.61.**
- **3D MHD model computes Shercliff-Hartmann liquid metal flows at Ha = 500.**
- **Core velocity flattens with Hartmann boundary layer thinning down to 0.2 mm.**
- **MHD pressure drop gradient evaluated at 0.45 MPa/m for Pb-17Li channels.**